Stockhunt Test Task — Walk-Forward + ML Results

Research universe: ~631 PIT S&P 500 constituents, 2008-01-02–2026-08-11. 7 of the 20 available indicators have rule-based trading strategies (see §2), spanning all four required families. Walk-forward: 5y in-sample / 1y out-of-sample / 1y rolling step, 14 windows, stitched OOS = 2013-01-01–2026-08-11 (final window is a partial ~7.5-month stub). Cost: 5 bps/side everywhere except the Phase 11 cost sweep. All numbers below are reproduced by `python run_all.py`.

1. Summary

- **Best strategy overall**: `combo__sma_rsi__or` (SMA 50/200 long-only OR RSI 21-day long-only), stitched OOS **Sharpe 0.89**, 95% bootstrap CI **[0.34, 1.48]**, net return 551%, max drawdown -40%, profitable on 89% of the 605 tickers it ever traded. Survives Benjamini-Hochberg correction, applied across the 21 walk-forward/OOS finalists — 87 configurations were explored in total, including 66 used for initial in-sample screening that were not retested (no OOS return series to bootstrap; see §9). Selection bias from that initial screening step is not itself corrected for.
- **Wiring up volatility and volume indicators paid off, not just a checkbox.** Bollinger mean-reversion (volatility family, zero rule-based coverage before this pass) actually *won the in-sample sweep outright* (Sharpe 0.75 vs. SMA's 0.74), and both it (0.68 OOS) and MFI mean-reversion (0.64 OOS, volume family) landed ahead of the original RSI baseline (0.58 OOS) walk-forward. That said, SMA still wins walk-forward OOS despite losing the in-sample sweep — a clean illustration of exactly why walk-forward validation is required rather than trusting an in-sample ranking (see §8).
- **Regime-filter, the previously-missing 4th combination type, works — for one of the two pairings tested.** Gating SMA with an ADX trending-regime filter improved it (0.78 vs. 0.77 ungated) — the classic rationale held. Gating RSI mean-reversion with a low-volatility-regime filter *hurt* it (0.49 vs. 0.58 ungated) — the classic "mean-reversion works better in calm markets" rationale did not hold here; RSI's edge in this dataset appears concentrated in sharp, higher-volatility reversals (2020, 2022), not quiet range-bound stretches, so filtering those episodes out removed edge rather than noise. Reported as a real, non-obvious negative result, not smoothed over.
- **The long-side-more-robust hypothesis survives walk-forward, sharply**: the best RSI long-short config scored Sharpe 0.50 in-sample; walk-forward OOS it collapsed to **-0.05** (95% CI [-0.57, 0.43], not distinguishable from zero). Most long-only lines held up (SMA 0.77, Bollinger 0.68, MFI 0.64, RSI 0.58, MACD 0.43 OOS) — but not every one: Stochastic was flat (0.03) and Donchian broke down (-0.88, the worst of any line tested, see §8).
- **ML does not beat the best rule-based Sharpe.** Best ML model (logistic regression) scored 0.76 OOS vs. the best rule-based combo's 0.89. Per the task brief, this is reported as a strong result for the rule-based approach, not hedged. LightGBM did *worse* than the logistic baseline (Sharpe 0.18).
- **Neither the best rule-based nor the best ML strategy beats simply holding the research universe equal-weight (Sharpe 0.99) or SPY (Sharpe 0.91) over this OOS period.** This is the single most important qualifier on every other result in this report — see §6 and §10.
- **Most long-only rule-based strategies — old and new alike — are 0.86–0.96 correlated with the equal-weight basket benchmark** (SMA, RSI, MACD, Bollinger, MFI, Stochastic). They are close to a filtered version of market beta, not an independent source of return. Two lines are clear exceptions: the regime-filtered RSI+low-vol combo (0.65) and Donchian breakout (0.64) — restricting trading to specific volatility regimes (or, for Donchian, simply performing badly) makes both meaningfully less beta-like, and both are also among the lower-Sharpe results in the study. The two ML models (-0.14, 0.03) and the RSI+Donchian AND combo (0.03) are the near-zero-correlation return streams in the study — not only the ML models.

2. Method

- **Data foundation** (~6-8 hours, built before the work in the rest of this report): PIT S&P 500 membership reconstruction, Yahoo price download and ticker mapping (renamed/delisted securities), corporate-action and OHLC-consistency validation, missing/corrupted price history investigation. Several of its decisions are directly load-bearing for the no-look-ahead and cost-realism claims made throughout this report: a fixed 2026-08-11 research cutoff so later Yahoo data can never leak in; NYSE-session-aware next-day return construction (`build_next_day_returns`), including correct handling of extended exchange closures (e.g. Hurricane Sandy, Oct 2012), not a naive calendar-day shift; known-corrupted tickers excluded by name (`EXCLUDED_PRICE_TICKERS`); extreme returns flagged and retained, never clipped; and a signal without a valid next-session return treated as untradable, never a fabricated fill. Validated with an initial 34-config in-sample sweep before any walk-forward/ML/combination work began.
- **Indicator coverage**: `indicators.py` computes 20 indicators spanning trend, momentum, volatility, and volume. Before this pass, only 3 had rule-based trading strategies (`sma_crossover`, `rsi_mean_reversion`, `donchian_breakout` — trend and momentum only). This pass adds 4 more, one genuinely representative of each remaining gap: `macd_crossover` (trend), `stochastic_crossover` (momentum), `bollinger_mean_reversion` (volatility — previously zero coverage), `mfi_mean_reversion` (volume — previously zero coverage). **7 of 20 now have rules, not all 20** — a prioritized, family-complete subset given time constraints, stated honestly rather than presented as full coverage. The remaining 13 indicators are still used as ML features (§5).

- Phase 6 shortlist (10 configs: 6 curated + 1 programmatically-selected best-in-sample-long-only pick per new family) was screened by **in-sample Sharpe from the 66-config sweep**, not from inside the walk-forward loop itself. Stated as a deliberate, time-constrained compromise.
- Walk-forward re-optimizes over each strategy family's *full* parameter grid every window, so shortlist entries sharing a (family, mode) collapse to one walk-forward line — 10 shortlist entries produce 8 distinct lines, not 10.
- Combinations (Phase 8): 3 fixed-parameter pairs (SMA+RSI, SMA+Donchian, RSI+Donchian) × {AND, OR, weighted-vote}, plus 2 regime-filter pairs (SMA gated by an ADX trending-regime signal; RSI mean-reversion gated by a low-volatility-regime signal) — the 4th combination type named in the task brief, previously missing. `combine_signals(method="regime_filter")` in `src/walkforward/combos.py`.
- ML (Phase 9, unchanged by this pass — it already used all 20 indicators' worth of feature columns regardless of which had rule-based strategies): 30 of 35 available indicator columns, label = sign(next-day return), purged (1-day) + embargoed (5-day) walk-forward CV aligned to the identical windows used everywhere else. No `train_test_split`/`KFold` anywhere. Trading rule: P(up)≥0.70 long, P(up)≤0.30 short, else flat.

3. Top strategies (rule-based, single)

line_id	Family	Sharpe	Net return	Max DD	% months profitable
`sma_crossover__long_only`	trend	0.770	352%	-37.8%	67.1%
`bollinger_mean_reversion__long_only`	volatility	0.682	389%	-43.5%	63.4%
`mfi_mean_reversion__long_only`	volume	0.635	335%	-46.3%	61.6%
`rsi_mean_reversion__long_only`	momentum	0.583	318%	-43.6%	59.8%
`macd_crossover__long_only`	trend	0.428	116%	-36.0%	62.2%
`stochastic_crossover__long_only`	momentum	0.030	-13%	-48.1%	56.1%
`rsi_mean_reversion__long_short`	momentum	-0.053	-14%	-33.5%	45.1%
`donchian_breakout__long_only`	trend	-0.881	-90%	-91.6%	44.5%

Every family now has a positive-Sharpe long-only representative except trend's second entry (MACD, still positive but weak) and momentum's second entry (Stochastic, essentially flat). Donchian remains the deliberately weak comparator and stayed weak out-of-sample. Stochastic crossover's best in-sample config was already negative (-0.13, see `results/shortlist.csv`) — the version shown here is its best walk-forward OOS run, still barely above zero. Not every indicator produces a working rule; reported as found, not cherry-picked.

4. Best combinations

line_id	Method	Sharpe	Net return	Max DD
`combo__sma_rsi__or`	OR	**0.888**	551%	-40.1%
`combo__sma_rsi__weighted_vote`	weighted (0.6 SMA / 0.4 RSI)	0.852	440%	-37.8%
`combo__sma_donchian__weighted_vote`	weighted (0.6 SMA / 0.4 Donchian)	0.852	440%	-37.8%
`combo__sma_donchian__or`	OR	0.816	399%	-37.8%
`combo__sma_adxtrend__regime_filter`	regime-filter (ADX-trend gate)	0.785	370%	-37.6%
`combo__rsi_donchian__weighted_vote`	weighted (0.6 RSI / 0.4 Donchian)	0.731	487%	-43.6%
`combo__sma_rsi__and`	AND	0.527	203%	-41.1%
`combo__rsi_lowvol__regime_filter`	regime-filter (low-vol gate)	0.488	203%	-48.2%
`combo__rsi_donchian__or`	OR	0.413	129%	-42.5%
`combo__rsi_donchian__and`	AND	-0.178	-5%	-7.0%
`combo__sma_donchian__and`	AND	-1.121	-93%	-94.7%

OR and weighted-vote both improved on their best individual leg in every pair; AND was worse than the better leg in every pair — requiring both signals to agree filtered out too many genuinely profitable trades. The ADX-trend regime filter on SMA (0.785) *beat* plain SMA (0.770) — turnover fell (fewer, better-timed trades) without giving up much upside, a genuine diversification-via-selectivity result. The low-vol regime filter on RSI (0.488) *underperformed* plain RSI (0.583) — see §1 for why. `combo__sma_donchian__or` matches `..._weighted_vote` almost exactly because Donchian is so rarely active that OR and a 0.6/0.4 vote resolve to nearly the same trades.

Combination parameters are fixed, not re-optimized per window: each leg's parameters come from its own walk-forward loop, but which legs to combine, the combination method, and any vote weights/regime thresholds are chosen once and reused unchanged across all 14 OOS windows — unlike single-strategy parameters, which do re-optimize each window (§2). A methodological limitation, not a gap in the four combination types tested.

5. Rule-based vs. ML head-to-head

line_id	Model	Sharpe	Net return	Max DD	Tickers traded	% profitable
`ml_logistic_regression`	Logistic Regression	0.757	1156%	-49.5%	29	51.7%
`ml_lightgbm`	LightGBM	0.177	17%	-56.1%	—	—
(best rule-based, for reference) `combo__sma_rsi__or`	—	0.888	551%	-40.1%	605	89.3%

The best rule-based Sharpe (0.89) beats the best ML Sharpe (0.76) — reported as a strong result for the rule-based approach, not hedged, per the task brief.

LightGBM underperformed its own simpler logistic baseline. Permutation feature importance shows why: logistic regression's top features are `price_sma_dist_50` and `rsi_14` — directly the same signals that won as standalone rules (SMA and RSI both being the walk-forward top performers), i.e. convergent evidence. LightGBM's top features are `cmf_20`, `volume_roc_20`, `adx_14` — volume-based indicators that did not independently win as rules anywhere in this study, consistent with it having found relationships that generalized poorly. Notably, neither model's top features are `bb_zscore_20_2.0` or `mfi_14`, despite Bollinger and MFI being the two next-best standalone rules found this pass — the ML models and the rule search surfaced *different* signal, not fully overlapping evidence. See `results/charts/ml_feature_importance.png`.

The logistic model's 0.76 Sharpe is fragile in a way its headline number doesn't show: only 15% of months were profitable, and it only ever took a position in 29 of the 631 tickers in the universe (avg. gross exposure 8.3%) — a threshold-triggered, low-frequency, concentrated strategy. Its equity curve (`results/charts/equity_curves.png`) is visibly a staircase of long flat stretches and a few large jumps (notably 2024), not a smooth compounding line.

Feature-representation ablation (`src/pipeline/run_ml_pit_percentile_ablation.py`, `results/ml_pit_percentile_ablation.csv`, not part of the primary result above): replacing raw indicator values with same-date cross-sectional percentile ranks (`build_ml_dataset(..., use_percentile_rank=True)` in `src/ml/dataset.py`) is a principled normalization — it makes a feature comparable across tickers regardless of absolute scale or a market-wide regime shift — but it broke the fixed 0.70/0.30 trading threshold rather than improving on it. Percentile ranks are uniform on [0,1] by construction, with no fat tails, so logistic regression's predicted P(up) compressed to **[0.454, 0.559]** — it never once crossed 0.70 across 88,928 first-fold predictions, i.e. the model made zero trades. LightGBM widened slightly to [0.292, 0.641] but still only fired once out of 88,928. The threshold was calibrated implicitly against the raw feature distribution; changing the feature representation without re-calibrating the threshold together with it silently disables the trading rule. Kept as a documented ablation rather than adopted as the primary result — recalibrating the threshold under the new distribution would mean abandoning the literal 0.70/0.30 framing, which was a deliberate choice (§2), not something to trade away for a normalization technique that, on this evidence, doesn't actually improve the outcome here.

6. Benchmarks

	Sharpe	Net return	Max DD
`Equal-weight universe basket`	**0.990**	827%	-40.3%
`SPY buy-and-hold`	0.915	564%	-33.7%
(best of this study, for reference) `combo__sma_rsi__or`	0.888	551%	-40.1%

Both benchmarks beat every single strategy tested in this project — old and new indicators alike — over the same stitched OOS window. This is the headline caveat on every result above — see §10.

7. Cost sensitivity

Best rule-based and best ML finalist re-run at 0/5/10/15/20/25 bps (0 bps used only here, per the hard constraint) — unchanged from before this pass since neither finalist changed:

Cost (bps)	`combo__sma_rsi__or` Sharpe	`ml_logistic_regression` Sharpe
`0`	0.914	0.772
`5 (baseline)`	0.888	0.757
`10`	0.863	0.743
`15`	0.838	0.728
`20`	0.813	0.713
`25`	0.788	0.699

Neither edge dies anywhere in this range. Chart: `results/charts/cost_sensitivity.png`.

8. Walk-forward efficiency and ranking stability

line	avg(OOS Sharpe / IS Sharpe)	avg rank corr (IS vs OOS)	% windows IS winner = OOS best
sma_crossover__long_only	1.50	0.11	21%
bollinger_mean_reversion__long_only	1.17	0.21	29%
mfi_mean_reversion__long_only	1.20	0.03	36%
rsi_mean_reversion__long_only	1.04	0.23	7%
macd_crossover__long_only	2.72	0.34	21%
stochastic_crossover__long_only	-5.29	0.27	36%
rsi_mean_reversion__long_short	-0.03	0.07	7%
donchian_breakout__long_only	n/a (IS Sharpe never positive)	0.17	0%

Bollinger beating SMA in-sample but losing to it out-of-sample (§1) is the clearest illustration in this dataset of why the task brief requires walk-forward rather than an in-sample ranking — an in-sample-only report would have shipped Bollinger as the headline strategy. Stochastic's -5.29 "efficiency" is an artifact of a near-zero in-sample Sharpe denominator, not a meaningful ratio — included for completeness, not implying a 5x blowup is real. Ranking stability is weak everywhere (rank correlations 0.03–0.34) — the parameter grids are small and per-window differences between candidates are mostly noise.

9. Robustness

Block-bootstrap Sharpe CIs and multiple-testing correction (1000 resamples, 20-day blocks; Benjamini-Hochberg at α=0.05 across the 21 walk-forward/OOS finalists — out of **87 configurations tested in total**: 66 in-sample sweep + 21 OOS finalists; the sweep is disclosed for context, not re-tested since it has no OOS return series to bootstrap):

13 of 21 finalists remain significant after correction, including the top result (`combo__sma_rsi__or`, p≈0, 95% CI [0.34, 1.48]), both new standalone strategies (Bollinger p=0.017, MFI p=0.016), the ADX-regime-filter combo (p=0.004), the low-vol-regime-filter combo despite its lower raw Sharpe (p=0.040), and the ML logistic model (p≈0). **Not significant**: MACD (p=0.070, just misses), Stochastic, both long-short lines, `ml_lightgbm`, and both Donchian-involving AND combos. Full table: `results/robustness_bootstrap.csv`.

Parameter-stability: the original sparse 6-point SMA sweep grid understated this — its nearest tested neighbor to the 50/200 winner was 30/100, a real gap (0.74 vs. 0.57) that couldn't rule out a local spike. A dense 5×5 local grid was built around each of the 5 finalists with genuine positive Sharpe (SMA, Bollinger, MFI, RSI, MACD — Donchian, Stochastic, and RSI long-short excluded as non-winners), varying each strategy's two structural parameters while holding the rest fixed at their winning values, in-sample, same methodology as the original sweep. **All five show a real plateau, not a spike**: SMA ranges 0.62–0.78 (spread 0.16) across the whole 5×5 neighborhood, MACD 0.26–0.33 (spread 0.065, the tightest of the five), RSI 0.46–0.77 (spread 0.31), Bollinger 0.46–0.81 (spread 0.35), MFI 0.52–0.71 (spread 0.19) — every cell in every grid stays positive, and each original winner sits inside its plateau rather than at an isolated peak (Bollinger's grid actually found a nearby point 0.06 Sharpe *better* than the original shortlisted pick — window=17/entry=-0.5 vs. window=20/entry=-1.5 — evidence the original coarse sweep simply hadn't sampled that neighborhood, not that the result is fragile). Charts: `results/charts/param_stability_{sma_crossover,macd_crossover,rsi_mean_reversion,bollinger_mean_reversion,mfi_mean_reversion}.png`.

Breadth: computed for all 21 tested strategies (`results/breadth_summary.csv`), not just the top finalist. `combo__sma_rsi__or` leads at 540/605 (89.3%); the next four are also combos (80–84%). Breadth and Sharpe don't always agree: `stochastic_crossover` has high breadth (79.7%) despite ~zero Sharpe (many small winners, no edge after costs), and `donchian_breakout`/`ml_lightgbm` show the opposite pattern — `donchian_breakout` is profitable on 48.7% of names yet has the worst Sharpe and drawdown in the study (a few catastrophic losers overwhelm many modest winners); `ml_lightgbm` is profitable on 61.9% of the (few) names it traded, consistent with its Sharpe now being weakly positive rather than negative (§5) — still the weaker of the two ML models by a wide margin. `ml_logistic_regression` remains the narrowest bet in the study: 15/29 (51.7%).

Survivorship bias, quantified (the PDF's explicit ask, not previously closed): re-running `combo__sma_rsi__or`'s exact stitched OOS signal, unchanged, restricted to only the 474 tickers that are S&P 500 constituents *today* (vs. the full 609-ticker PIT-reconstructed universe the signal actually traded) — same window, same cost — **survivorship bias overstates the Sharpe ratio by +0.086 (0.891 → 0.977) and net return by +110 percentage points (555% → 665%)**. Direction: using today's list instead of the correct PIT list makes a strategy look better than it is, as expected — a non-PIT-aware backtest on this data would have overstated the edge by roughly 10% of the Sharpe itself. `results/survivorship_bias_quantification.csv`.

10. Where results are weakest

- **No strategy in this study beat its own benchmark**, old or new. The equal-weight basket (0.99 Sharpe) and SPY (0.91) both beat the best combo (0.89) and the best ML model (0.76).

- **The rule-based edge is mostly beta, not alpha, and this held even after adding 4 new indicator families.** The six strongest long-only lines tested — SMA, RSI, MACD, Bollinger, MFI, Stochastic — are 0.86–0.96 correlated with the equal-weight basket (§1/§9, `correlation_matrix.csv`). Two lines are exceptions: the low-vol-regime-filtered RSI combo (0.65) and Donchian breakout (0.64) — both also among the lower-Sharpe results, so the diversification and the underperformance appear to be the same phenomenon (trading a narrower slice of market conditions, in Donchian's case a mostly-broken one).
- **7 of 20 indicators have rules, not all 20.** Trend and momentum each have a second, weaker representative (MACD, Stochastic); volatility and volume each have exactly one (Bollinger, MFI). The remaining 13 are ML features only. Stated explicitly rather than left implicit.
- **In-sample-screened shortlist** (§2): the shortlist that entered walk-forward was chosen by in-sample Sharpe. Bollinger's in-sample win over SMA (§8) shows this ranking is not reliable — a strategy screened only inside the walk-forward loop from the start might have produced a different top set.
- **ML logistic result is concentrated**: 29 tickers, 15% of months profitable, staircase equity curve.
- **Regime-filter's second pairing (RSI + low-volatility gate) underperformed the ungated strategy** — included and explained (§1, §4), not dropped for looking bad. Only 2 regime-filter pairings were tested; a broader sweep of gate thresholds or gate/base-strategy pairings was out of scope for this pass.
- **Turnover-at-roll-date artifact**: `run_backtest` computes turnover from a weight-matrix pivot of only the rows it's given, so day 1 of every stitched OOS segment looks like a from-flat entry rather than a continuation of the prior window's book — a small, documented mechanical overstatement of turnover/cost at each of the 13 roll dates. Not a look-ahead leak.
- **Weak ranking stability** (§8): the walk-forward re-optimization step is not clearly finding a durable signal above noise at this grid resolution.
- The two stateful mean-reversion state machines with the slowest per-call runtime (`_rsi_target_state_long_only/_long_short`, reused by MFI; the Bollinger equivalent) are plain per-ticker Python loops, not vectorized — functionally correct (see `tests/test_no_lookahead.py` and the smoke tests) but a real performance gap, not just a style choice: the dense parameter-stability grids for Bollinger and MFI took most of this pass's total runtime for exactly this reason.
- **Cross-sectional PIT percentile normalization was attempted, not skipped** (§5 ablation) — tried, diagnosed as breaking the fixed confidence threshold, not adopted. Still genuinely not attempted: SHAP feature importance (permutation importance used instead); long-short combos; regime-filter pairings beyond the 2 tested; wiring the remaining 13 indicators as rules; PCA/SHAP/Boruta-driven feature reduction ahead of model training (5 redundant columns were dropped by manual inspection instead, see `src/ml/dataset.py`); ML hyperparameter tuning (both models use fixed, untuned defaults); a re-calibrated threshold that would make the percentile-normalized features usable (e.g. per-window quantile-based cutoffs instead of a fixed value).

Appendix: reproducing these numbers

`python run_all.py` reproduces every number in this report from a genuinely clean clone — no manual setup required, including the data foundation (membership reconstruction, price download, PIT dataset, features, in-sample sweep). Budget well over an hour for the network-bound data download alone; `python run_all.py --skip-data` re-runs only the research build if that foundation already exists on disk. See `README.md` for the full phase table and column glossary.

Correction applied during review: `spy_buy_and_hold` (`src/evaluate/benchmarks.py`) originally built SPY's return series with `pct_change()` (return on day t labeled as the t-1 -> t move), one trading session out of alignment with every strategy and the equal-weight basket, both of which use the project's t -> t+1 convention (signal on day t, return realized t -> t+1). This didn't materially move SPY's own Sharpe/net-return/max-DD (§6) but did corrupt SPY's correlation against every other return stream. Fixed by reusing `build_next_day_returns` for SPY, identically to the equal-weight basket; `benchmark_results.csv` and `correlation_matrix.csv` were regenerated after the fix.